

Trung Le

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Skills

Core Languages: C/C++, GPU shader programming

Secondary Languages: Python, Rust, Swift, Javascript

Graphics APIs: Vulkan, WebGL, OpenGL

Tools/Engines: Unreal, Unity, Houdini, Blender

Experience

SENIOR GRAPHICS ENGINEER, BAD ROBOT GAMES - JAN 2024 - PRESENT

Led graphics engineering efforts as sole graphics engineer. Worked closely with tech art, lighting, and VFX teams to provide support, knowledge sharing, and building custom tooling structures – notably a networked modular lighting system for dynamic time of day and weathers. Managed performance optimization targeting 60FPS across console and PC platforms, focusing on GPU and memory usage. Coordinated with QAs, principal engineers, and automation team to establish performance workflows.

GRAPHICS ENGINEER, DEVIATION GAMES - MAY 2021 - JAN 2024

Implemented real-time signed distance field ray marching for animated characters in Unreal. This feature uses skinned point clouds as an underlying data structure. Collaborated with the R&D team and gameplay engineers to integrate for enemy combat system. Worked on various rendering features, including portal rendering and GPU tessellation.

ML ENGINEER, GOOGLE STADIA - OCTOBER 2019 - MAY 2021

Advocated for and applied neural rendering research toward solving real-time graphics problems for video games. My focuses were AI art generation and high fidelity fur rendering. Worked on generating custom 3D datasets using Houdini and Blender, training generative machine learning models with Google infrastructure, parameter tuning, and interfacing novel workflows with artists.

AI PROGRAMMER, ROCKSTAR GAMES - JUL 2017 - OCT 2019

Analyzed and optimized performance for AI systems in Red Dead Redemption II, Rockstar's premier narrative-driven, open world video game title with deep AI interactions. Specifically worked to improve the C++ game code for LOD management of AI physics, weapon and combat systems, transport systems, and pathfinding systems for PS4 and Xbox. Worked on general AI and animation optimization for several specific scripted in-game missions as well as miscellaneous performance bugs in the final weeks before release.

TOOLS PROGRAMMER INTERN, EPIC GAMES - JUNE 2016 - SEP 2016

Implemented VR editing tools for Unreal Engine 4. The tools include terrain editing, virtual keyboards, and foliage painting for level editing. These tools were introduced in the Unreal 4.13 and 4.14 updates.

FIRMWARE ENGINEER, JAWBONE – 2012-2014

Developed infrastructure and applications for the UP3 fitness wristband on ARM Cortex and iOS platforms. This included the BTLE protocol, authentication and encryption between devices and the mobile app, activity classification collection tools, peripheral drivers, USB interface, and UX.

RESEARCH ASSISTANT, UW SENSOR LAB+INTEL LAB – 2010-2012

Designed a GUI with Python QT for the systems used in Wireless Resonant Energy Link (WREL) research. The software supports data collection, data visualization, wireless control, and power diagnostics. Over the years, this software has been extended for use at the startup company Wibotic and in other research ventures.

Projects

(please see more at www.trungtuanle.com for a complete portfolio)

Education

University of Pennsylvania, PA – MSE Computer Graphics, May 2017

University of Washington, WA – BS Electrical Engineering, June 2012